Aleksa Vucak

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Education

University of Windsor — Sep 2023 – Aug 2027
Bachelor of Science (Honours) - BS, Computer Science with Software Engineering Co-op — *Windsor, ON*

Experience

Software Engineer Co-op — Jan 2026 – Present
Stellantis (NYSE: STLA) — *Windsor, ON (Hybrid)*

- Engineered **ImpedanceZ** for Stellantis' **OTA** team using **React**, **TypeScript**, **Three.js**, **FastAPI**, **SQLite**, and **C++** to unify 3D circuit design, simulation, and hardware control, reducing PCB prototype costs by **$1,050**/cycle
- Integrated **C++** firmware on **Arduino/PlatformIO** with **pyserial**, **FFT-based analysis**, and **Python** solvers to simulate circuitry, validate physical tests, streamline signal processing, and reduce turnaround time by **56%**
- Developed a **Python** clustering and record-linkage pipeline that auto-tuned time-gap thresholds, processed torque sequences, and assigned BVINs, reducing human intervention by **99%** using **Selenium** and **PyTest**

Undergraduate Teaching Assistant — Sep 2025 – Present
University of Windsor — *Windsor, ON (Hybrid)*

- Led a weekly Software Development lab for **37 students**, teaching **Java**, **OOP**, and **MySQL** through live coding, debugging demos, and guided practice with classes, inheritance, collections, queries, joins, and schema design

Machine Learning Engineer Co-op — Apr 2025 – Aug 2025
Stellantis (NYSE: STLA) — *Windsor, ON (On-site)*

- Built a battery anomaly detection pipeline in **Python** using **pandas**, **NumPy**, **scikit-learn**, and **XGBoost** to extract BMS features and improve precision at target recall by **21%** on tested cells and packs
- Designed **SoC-aware calibration** and validation-driven thresholds using **Python** and precision-recall analysis, reducing false positive alerts by **32%** while preserving recall and preventing evaluation leakage
- Streamlined experiment tracking and engineer-ready reporting in **Python**, reducing triage time per log by **60%** across testing/validation workflows

Computer Vision Engineer Intern — Sep 2024 – Dec 2024
Glendor, Inc — *Draper, UT (Remote)*

- Created a medical de-identification pipeline in **Python** using **OpenCV-based** blur and detection algorithms to process **2000+** medical images, achieving **92%** accuracy in identifying sensitive regions while preserving privacy
- Automated output generation and asset organization with **Python** scripts, reducing manual intervention by **96%** through metadata tagging, file renaming, and structured folder storage

Projects

TradeCore | *Next.js, TypeScript, FastAPI, Rust, PostgreSQL, Redis, Celery, scikit-learn, Docker, Git*

- Architected a full-stack **crypto strategy research and paper-trading platform** that ingests historical market data, executes user-scoped backtests, and surfaces trade-level analytics, equity curves, and benchmark comparisons
- Built asynchronous market-data, validation, and **ML-regime** workflows with **Celery**, **Redis**, and **WebSockets**, enabling walk-forward analysis, real-time job tracking, and strategy evaluation across backtesting pipelines

ReflectAI | *TypeScript, React Native, Expo, FastAPI, PostgreSQL, Redis, Celery, Docker, Git*

- Launched an **AI journaling mobile app** that delivers daily reflection prompts, supports typed and voice-based entries, and generates weekly insight summaries with recurring themes, patterns, and progress tracking
- Designed reminder, streak, and entry-history flows to support consistent journaling habits and long-term reflection

GameFlow | *Python, R, XGBoost, DuckDB, Parquet, Streamlit, Quarto, Git*

- Created an **NBA analytics platform** that models in-game win probability from play-by-play data, ranks the biggest swing plays by win-probability, and compares calibrated **Python** and **R** baseline models through **Quarto**

Technical Skills

Languages: Python, Java, C, C++ (CUDA), Rust, SQL, TypeScript, Golang, R, Bash/Shell, HTML, CSS, PHP
Frameworks/Libraries: FastAPI, React, Next.js, Node.js, Spring Boot, Express.js, NestJS, React Native, Expo, scikit-learn, PyTorch, TensorFlow, pandas, NumPy, SciPy, PySpark, SQLAlchemy, OpenCV, Matplotlib
Tools/Platforms: Git, GitHub, Linux/Unix, Docker, AWS/Azure, CI/CD, Kubernetes, Redis, Celery, Jupyter, Postman, Kafka, Databricks, Alembic, PyTest, JUnit, Selenium, Jira, Confluence, OpenAPI, Power BI, Tableau